

SKILLSHEET

Order of operations

If an expression contains more than one operation, the calculations must be performed in the following order:

1. brackets
2. multiplication and division (from left to right)
3. addition and subtraction (from left to right).

WORKED EXAMPLE

Find the value of each of the following, using the order of operations rules.

a $12 \div 2 + 4 \times 5$

b $15 + (6 + 4) \times 7 \div 10$

THINK

- a**
- 1 Write the question.
 - 2 There are three operations to be performed: division, addition and multiplication. According to the order of operations rules, multiplication and division must be done before the addition, from left to right. So, perform the division first, followed by the multiplication and finally the addition.
- b**
- 1 Write the question.
 - 2 The calculation in the brackets must be done first, followed by multiplication and division. These are done in order of appearance (that is, from left to right), so in this case multiplication must be done before division. Addition is done last.

WRITE

a $12 \div 2 + 4 \times 5$
 $= 6 + 4 \times 5$
 $= 6 + 20$
 $= 26$

b $15 + (6 + 4) \times 7 \div 10$
 $= 15 + 10 \times 7 \div 10$
 $= 15 + 70 \div 10$
 $= 15 + 7$
 $= 22$

Try these

Find the value of each of the following, using the order of operations rules.

1 $23 - (15 - 6) \div 3$

$= 23 - \dots \div 3$

$= 23 - \dots$

$= \dots$

2 $40 - 8 \times 4 + 2$

$= 40 - \dots + 2$

$= \dots + 2$

$= \dots$

3 $56 - 49 \div 7$

$= 56 - \dots$

$= \dots$

4 $12 \div (54 \div 9) + 3$

$= 12 \div \dots + \dots$

$= \dots + \dots$

$= \dots$

5 $14 - (16 - 5) \div 11$

$= \dots$

$= \dots$

$= \dots$

6 $18 + 6 \times 8 \div 8 \times 3$

$= \dots$

$= \dots$

$= \dots$

$= \dots$

7 $10 + 8 \times 2 \div 8$

=
=
=

8 $144 \div 12 + 6 \times 2 - 5$

=
=
=
=

9 $10 + 6 \times 3 \div 6 \times 8$

=
=
=
=

10 $72 - 64 \div 8$

=
=

11 $18 - 7 \times 2 + 9$

=
=
=

12 $2 \times (63 \div 9) + 3 \times 2$

=
=
=
=

13 $12 + 12 \times 2 \div 6$

=
=
=

14 $21 \div (4 + 3) \times 9$

=
=
=

15 $6 + 6 \times 4 + 8$

=
=
=

16 $(6 + 3) \times (7 + 2)$

=
=
=

17 $11 \times 8 + 9 \times 3$

=
=
=

18 $7 + 3 \times (4 + 3)$

=
=
=

19 $8 + 4 \times (6 + 3 \times 2)$

=
=
=
=

20 $(7 + 2) \div (12 - 3)$

=
=
=

SKILLSHEET — ANSWERS

SKILLSHEET

Order of operations

- | | |
|--------|-------|
| 1 20 | 2 10 |
| 3 49 | 4 5 |
| 5 13 | 6 36 |
| 7 12 | 8 19 |
| 9 34 | 10 64 |
| 11 13 | 12 20 |
| 13 16 | 14 27 |
| 15 38 | 16 81 |
| 17 115 | 18 28 |
| 19 56 | 20 1 |